

Sasidhar Guthi
Data Engineer
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PROFESSIONAL SUMMARY

Data Engineer with around 5 years of experience designing and optimizing large-scale, cloud-native data platforms across healthcare, retail, and financial domains. Skilled in building high-performance ETL and streaming pipelines using AWS Glue, Lambda, EMR, and Kinesis, integrating Snowflake and Spark for analytics and AI-driven automation. Adept at implementing retrieval-augmented generation (RAG) frameworks, real-time processing systems, and CI/CD automation to deliver secure, scalable, and intelligent data solutions.

WORK EXPERIENCE

Client: Walmart | Vizio, Addison, TX

Apr 2024 - Present

Data Engineer

Functional Role Details:

End-to-end data owner for the Vizio Mobile App, leading data strategy, telemetry governance and business intelligence

- Automated end-to-end financial data workflows using **Python**, **Scikit-Learn** and **Boto3**, enabling parallel processing, efficient serialization and seamless AWS integration.
- Built and launched real-time diagnostic dashboards for the Vizio Mobile App by integrating streaming pipelines, reducing latency detection time by 40% and enabling early identification of stability regressions.
- Leveraged **DynamoDB** as a metadata store for pipeline configurations and job tracking, streamlining orchestration with **Step Functions** and using automated lifecycle management, saving hours of manual maintenance each week.
- Independently built and deployed a full-stack event validation platform (React.js frontend, FastAPI backend running on AWS Lambda with an API Gateway handler) to validate mobile telemetry for each release, improving data accuracy and accelerating release cycles by reducing manual QA effort.
- Enabled JSON message exchange between **REST APIs** and stream processors (Kafka, Kinesis), powering end-to-end real-time analytics pipelines.
- Conducted load testing and latency profiling of **REST endpoints** using tools like JMeter and Locust, tuning performance bottlenecks in high-throughput environments.
- Built fault-tolerant ETL pipelines with retry logic, checkpointing and real-time streaming integration, enabling resilient data workflows and unified analytics across batch and live data sources.
- Deployed containerized services using **Docker** and **Kubernetes**, leveraging auto-scaling and service discovery for fault-tolerant data processing applications.
- Improved collaboration and deployment workflows by implementing merge request pipelines in **GitLab**, automating test suites and ensuring high-quality code delivery with reduced manual intervention.
- Diagnosed and improved ETL job efficiency by profiling **AWS Glue** performance metrics, identifying bottlenecks and optimizing resource allocation to reduce overall runtime and cost.
- Collaborated on designing and implementing a **Retrieval-Augmented Generation (RAG)** framework to support an **AI-driven NLP chatbot** capable of processing natural language inputs and generating SQL queries dynamically.
- Implemented **vector-based similarity checks** to connect the NLP processing flow with the RAG retriever, improving contextual accuracy and reducing irrelevant model calls.

Client: Merck Pharma, Charlotte, NC

Jun 2023 - Apr 2024

Data Engineer

Functional Role Details:

Led end-to-end healthcare data engineering, building real-time pipelines, optimizing Snowflake workloads, and ensuring HIPAA-compliant data access and reporting.

- Developed scalable data integration pipelines using AWS Glue and AWS Data Pipeline to streamline ingestion, transformation, and delivery of healthcare data across distributed systems and EHR platforms.
- Performed high-performance analytics on large-scale healthcare datasets using Amazon EMR with Apache Spark, accelerating clinical reporting, claims analysis, and population health insights.
- Created efficient data processing scripts in Python using Pandas and NumPy for healthcare data modeling and implemented Pytest to maintain code reliability across mission-critical workflows.

- Automated complex data workflows with Apache Airflow on Amazon MWAA, building resilient, scheduled pipelines to process sensitive patient, claims, and provider data.
- Ensured secure and reliable data access between healthcare applications and relational databases using **AWS Lambda** and **Boto3** integrations with **Amazon RDS** and **Amazon Aurora**, minimizing latency and supporting seamless system integration.
- Implemented real-time event processing with **Amazon Kinesis Data Streams** and **Kinesis Data Analytics**, enabling continuous monitoring and live data updates for operational dashboards.
- Built CI/CD automation pipelines using **GitHub Actions**, enforcing code quality through testing, linting and deployment hooks triggered on every code commit or pull request.
- Led agile execution of data platform initiatives using **Scrum methodologies**, managing sprint ceremonies, backlog prioritization and stakeholder engagement to deliver high-impact, iterative solutions.

Client: Infosys, Hyderabad, India

Sep 2019 - Aug 2021

Data Engineer

Functional Role Details:

Built scalable, event-driven data pipelines using AWS Lambda, Glue, Step Functions, and Spark, automating ingestion, transformation, and real-time analytics for high-volume trading systems.

- Spearheaded the development of end-to-end data automation workflows within financial systems by utilizing **AWS Lambda**, **AWS Glue** and **Step Functions**, driving high-performance ingestion, cleansing and transformation pipelines.
- Streamlined serverless deployment operations by automating code releases from **Amazon S3** to **AWS Lambda** and set up proactive health checks and metric dashboards with **CloudWatch** for real-time observability.
- Built asynchronous messaging and event-handling systems using **Lambda**, **SQS** and **Step Functions**, enabling reliable workflow orchestration and efficient data state tracking in production-grade finance environments.
- Directed real-time and batch data processing initiatives leveraging **Scala**, **Apache Spark**, **Kafka** and **Spark Streaming**, optimizing system throughput and latency in capital markets and trade data pipelines.
- Applied high-efficiency data serialization formats such as **Parquet** and **ORC** alongside **PySpark**, increasing pipeline throughput and facilitating advanced analytics across structured and semi-structured financial datasets.
- Oversaw the refinement of real-time analytics systems, integrating **Spark Streaming** with event-based pipelines to deliver timely insights, fraud detection and trend analysis for high-stakes financial decision-making.

TECHNICAL SKILLS

Programming Languages: Python, PySpark, SQL, Scala

Web Frameworks / APIs: FastAPI, Flask, Django

Databases (Relational): PostgreSQL, SQL Server, Oracle

Cloud Data Warehousing: AWS Redshift, AWS RDS, Azure Synapse Analytics, Google BigQuery

ETL & Data Integration Tools: AWS Glue, Azure Data Factory, Google Cloud Dataflow, Talend

Streaming & Real-Time Processing: AWS Kinesis, AWS Firehose, Kafka, Azure Stream Analytics

Big Data Ecosystem: Apache Spark, HDFS, Hive, Pig, YARN, MapReduce, Databricks

Business Intelligence & Visualization: AWS Athena, QuickSight, Power BI, Looker, Tableau

Cloud Platforms: AWS, Azure, GCP

Cloud Services: EC2, S3, IAM, VPC, Cloud SQL, Google App Engine, Azure Virtual Network, Azure Blob Storage, Azure Files

Infrastructure as Code (IaC): Terraform, AWS CloudFormation, ARM Templates

Orchestration Tools: Apache Airflow, Dagster, Prefect, Step Functions

Serverless & Event-Driven Computing: AWS Lambda, SNS, SQS, Azure Functions, Google Cloud Functions

Containerization & Deployment: Docker, Kubernetes, AWS ECS, AWS Fargate, Google Cloud Run

CI/CD & DevOps Tools: GitHub, Bitbucket, AWS CodeBuild, GitHub Actions, SVN

Machine Learning Platforms: AWS SageMaker, Azure AutoML, ML Studio, Vertex AI

Prompt Engineering / LLMs: Prompt Engineering, Large Language Models (LLMs)

EDUCATION

Bachelors in Electrical and Electronics Engineering

Andhra University

Aug 2015 - Jul 2019

Masters in Data Science

University of Maryland Baltimore County

Aug 2021 - May 2023